

ORIGINAL RESEARCH

Research on airport baggage anomaly retention detection technology based on machine vision, edge computing, and blockchain

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Abstract

Airport checked luggage entails specific requirements for speed, stability, and reliability. The issue of abnormal retention of checked luggage presents a significant challenge to aviation safety and transportation efficiency. Traditional luggage monitoring systems exhibit limitations in terms of accuracy and timeliness. To address this challenge, this paper proposes a real-time detection and alerting of luggage anomaly retention based on the YOLOv5 object detection model, leveraging visual algorithms. By eliminating cloud servers and deploying multiple edge servers to establish a private chain, images of anomalously retained luggage are encrypted and stored on the chain. Data users can verify the authenticity of accessed images through anti-tampering algorithms, ensuring the security of data transmission and storage. The deployment of edge computing servers can significantly reduce algorithm latency and enhance real-time performance. This solution employs computer vision technology and an edge computing framework to address the speed and stability of checked luggage transportation. Furthermore, blockchain technology greatly enhances system security during operation. A model trained on a sample set of 4600 images achieved a luggage recognition rate of 96.9% and an anomaly detection rate of 95.8% in simulated test videos.

1 | INTRODUCTION

Information security and stability in airport systems have always been major areas of concern for airport management [1]. The current checked luggage handling system upholds high standards in terms of efficiency, security, and stability [2, 3]. The primary method of luggage tracking at airports involves the utilization of radio frequency identification (RFID) technology to monitor luggage at various points [4–6]. However, accurately detecting abnormal luggage statuses in the intervals between two RFID scanning devices remains a challenge. The comprehensive detection of luggage abnormalities, along with the secure transmission, storage, and statistical analysis of this abnormal information, has become an urgent necessity for airports. With the advancement of artificial intelligence technology, research on computer vision based on deep learning [7–9] has matured significantly in object detection technology and is widely implemented in the transportation sector [10, 11].

In the hardware field, edge computing aims to bring computing resources and data storage closer to the data source and the end user, in order to reduce the latency of data transmission over the network, and to improve the system's response speed and performance. Edge computing emphasizes processing data closer to where it is generated and used, rather than relying on traditional centralized cloud computing architectures [12, 13]. The characteristic of edge computing is its ability to facilitate the processing of sensitive data locally, reducing the need to transmit data to the cloud, thereby enhancing data privacy and security [14, 15].

Blockchain technology is a form of distributed ledger technology that records and verifies transaction data in a decentralized and tamper-proof manner. Originally designed for Bitcoin, this technology has evolved over time to become a more widely applicable technology, spanning various fields such as finance, supply chain, healthcare, and the Internet of Things (IoT) [16, 17].

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In the application of computer vision and blockchain technology, Prince Waqas Khan and others [18] introduced blockchain technology into the closed-circuit television (CCTV) systems of smart cities. To ensure the security of video technology and personal privacy, the characteristics of blockchain's distributed ledger are utilized to store image information. This technology plays a significant role in the secure storage of video data. Gallo, P. and others [19] introduced a blockchain architecture system called BlockSee on the foundation of smart city CCTV systems. This system employs smart contract technology to grant different levels of video access rights to various personnel, such as citizens, police officers, etc. This technology not only ensures the security of citizens' privacy but also brings convenience to administrators. Selvi, C. and others [20] extensively reviewed the current research status of blockchain and computer vision regarding data security, data integrity, and related aspects. Mainstream research results are as follows: Blockchain can provide security guarantees for data security in computer vision systems; the contract function of blockchain can provide guarantees for data sharing; blockchain can improve the efficiency of distributed training for models. Therefore, the application of blockchain in the field of computer vision mainly focuses on data security and the application of smart contracts. In high-security public places like airports, especially in controlled areas such as checked luggage, the security requirements are relatively high. Combining blockchain technology with computer vision systems can effectively enhance the security of data transmission and storage in existing CCTV systems.

In the application of edge computing and blockchain, Gill, S. S. in the article [21], introduced the computing paradigms related to edge computing and their value in IoT applications. The research introduced blockchain technology for security and incorporated the concept of quantum computing for computational speed. This study has significant implications for the integration of edge computing and blockchain technology, providing valuable insights into enhancing security and computational efficiency. Liu, H. and others [22] proposed encrypting data based on blockchain and setting up smart contracts to ensure secure communication between cars. They further suggested publishing data to onboard mobile edge computing servers and cloud computing servers. This demonstrates the practical value of integrating blockchain with an edge computing. In Cheng, G3's research [23], it is elucidated that edge computing servers in IoT systems are susceptible to attacks. However, technologies such as mutual authentication using blockchain can elevate the security level between edge computing servers and IoT devices, as well as between edge computing servers themselves, thereby preventing intrusion. In summary, there are security threats present between edge devices, and blockchain technology can effectively mitigate these threats.

In the existing field of airport baggage anomaly detection, Singh, A. and others [24] have iterated upon traditional baggage RFID technology. They proposed an IoT technology based on smart RFID tags and cloud server processing to detect anomalies such as baggage loss and mishandling. This technology has relatively good effects on baggage tracking. Shobayo and others aimed [25] to accurately locate lost luggage by implementing

location tracking using GSM/GPS (global system for mobile communications / global positioning system) modules. This approach helps prevent situations where luggage is lost or mis-routed and cannot be located. Siddique, A. and others [26] have already utilized convolutional neural network (CNN)-based techniques to track multiple targets, including passengers and luggage, across different cameras. This lays the foundation for subsequent algorithms implementing luggage anomaly detection functionalities. Wang, F. and others [27] have integrated blockchain technology with edge computing to store relevant information from the RFID identification process of luggage. This integration ensures the security of information while detecting luggage anomalies and also possesses decentralized characteristics. Research in luggage anomaly detection is trending towards the direction of artificial intelligence, blockchain, and edge computing. These technologies contribute to improvements in timeliness, security, and other aspects of luggage anomaly detection processes.

To detect instances where airport checked luggage is anomalously retained during the handling process, leading to situations where the luggage cannot be properly loaded onto the airplane, the proposed technical solution in this paper will combine computer vision technology, edge computing servers, and blockchain technology for real-time monitoring of checked luggage in airport handling equipment.

The combination of computer vision technology with edge computing framework allows algorithms to be executed close to the data source, enabling faster response times for timely detection and processing of anomalous retention of luggage, thereby enhancing security and efficiency. Data is processed and analysed locally, allowing sensitive information to be handled offline, reducing the risk of privacy breaches during data transmission. Edge computing can process data locally and only transmit key information to central servers, reducing the demand on network bandwidth and lowering transmission latency. Introducing blockchain technology further enhances the technical solution by providing a highly secure method for data storage and transmission. Information regarding anomalously retained luggage needs to be securely stored to prevent data tampering or deletion. The immutability and decentralized nature of blockchain ensure the security of information about abnormal luggage, protecting it from hacker attacks and data breaches during transmission and storage. Additionally, data on the blockchain maintains a complete history record and timestamps for every operation and transaction, which is valuable for tracking and auditing abnormal luggage, helping regulatory authorities better understand the origin and handling of such cases. As blockchain data is distributed and each block contains the hash value of the previous block, it becomes impossible for anyone to tamper with the data stored on the blockchain. This ensures the integrity of information about abnormal luggage, as any attempt to modify the data will be immediately detected.

The main contributions are as follows:

- Proposing a system architecture: Implementing comprehensive surveillance of the conveyor belt with CCTV systems. Deploying edge computing servers in onsite network cabinets

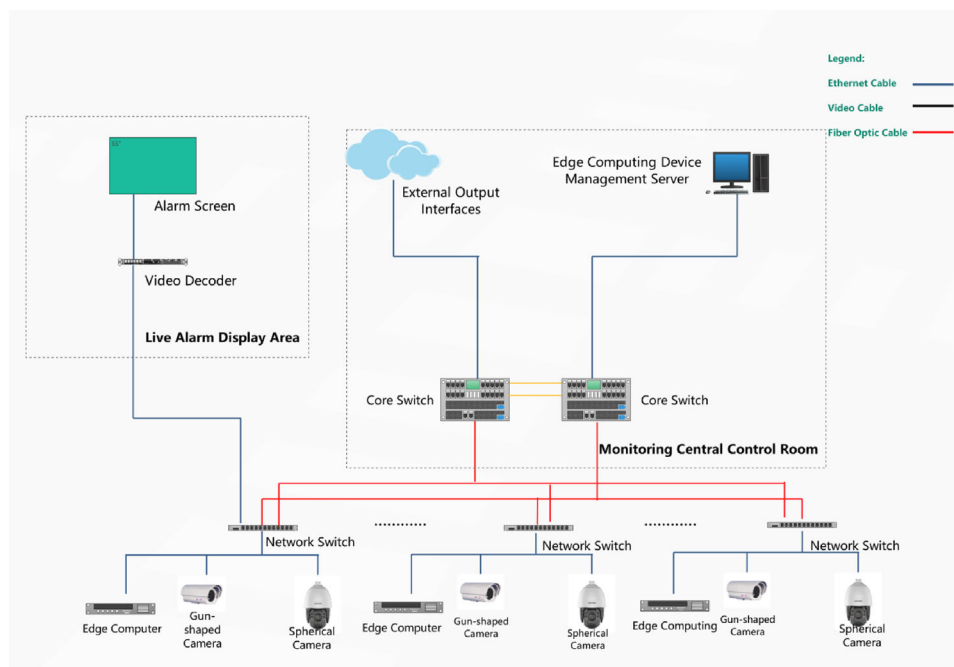


FIGURE 1 System and network architecture diagram.

for loading visual processing models, processing algorithms, and transmitting alarm results, etc., to reduce computational delay. Ensuring data security through encryption and distributed storage by deploying multiple edge devices to form a private blockchain.

- Utilizing a vision model trained with You Only Look Once (YOLOv5) [28] to detect luggage targets. Designing an algorithm to monitor and alert for luggage that falls off the conveyor belt.
- Encrypting abnormal luggage and storing it on edge devices. Encoding information such as luggage images, luggage types, timestamps, and camera numbers into hashes for tamper-proof verification on the blockchain, enhances the overall security of the technical solution and the confidentiality of data applications.
- Conducting experimental verification through testing and comparing it with various system solutions to validate the advantages of the proposed technical solution in complex areas, coverage, data security, and algorithm timeliness. Prospects and explorations for the future application of blockchain technology in this technical solution are also discussed.

The rest of this paper is organized as follows: Section 2 introduces the system hardware connections and network architecture adopted in this paper, clarifying the system's hardware and architecture. Section 3 elaborates on the application methods and effects of computer vision technology algorithms and edge computing devices in the scheme. Section 4 discusses the application of blockchain distributed storage and edge computing devices in the system. Section 5 provides interpretation and analysis of the tests. Finally, Section 6 concludes the paper

and explores prospects for the future application of blockchain technology in this system.

2 | NETWORK ARCHITECTURE DESIGN

With the development of artificial intelligence technology, an increasing number of object detection frameworks are emerging in the field of computer vision [29]. However, this has brought about issues related to insufficient computing power and timeliness. In the past few years, cloud servers have been the mainstream solution for providing powerful computing power. With the increase in devices and algorithm types, cloud servers have significant advantages. However, in work scenarios with strict network security requirements, like airports, the advantages of edge computing devices become more pronounced. Edge computing servers, being closer to the data source, reduce the distance of data transmission, typically resulting in lower latency. Additionally, because edge computing servers process data near the devices, they can handle sensitive information locally and form a private blockchain for distributed storage, enhancing privacy and security.

As shown in Figure 1, the entire system consists of three main components: the live alarm display area, the monitoring central control room, and on site equipment. In the live alarm display area, the primary setup includes alarm screens and video decoders, which are used to receive and display images of abnormal luggage to on-site personnel, enabling them to promptly handle detained luggage and enhance efficiency. The live alarm display area mainly comprises the edge computing device management server and the core switch. The edge

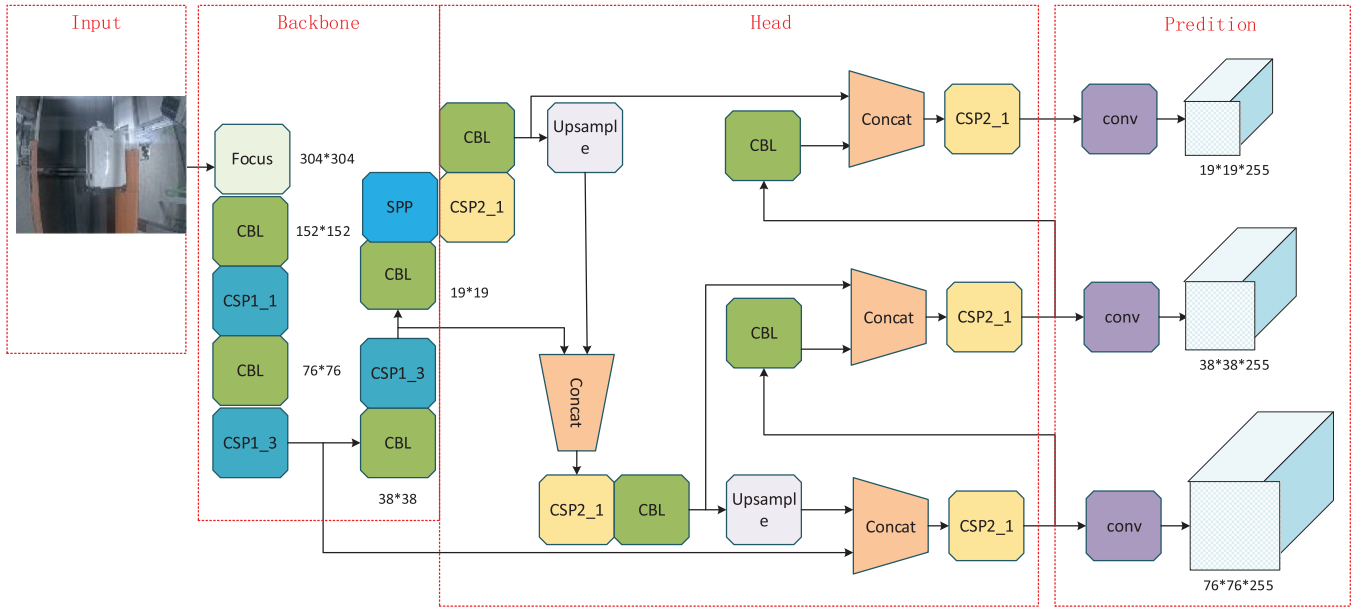


FIGURE 2 YOLOv5 network architecture diagram.

computing device management server is responsible for supervising and maintaining each edge computing device on-site to ensure their proper operation. Additionally, two core switches serve as backups to prevent system crashes in case of device failures. On-site equipment primarily consists of edge computing devices and network cameras. A single edge computing device can process video feeds from multiple network cameras (the specific number depends on factors such as algorithm functionality and video resolution). Multiple edge computing devices are deployed on-site, each connected to other edge computing devices and the edge computing device management server within the same LAN, forming a private chain. The main functions of the edge computing devices include loading vision models for inference and detecting luggage targets, loading algorithms to detect abnormal detained luggage, outputting images of abnormal luggage to alarm screens, and encrypting and storing images of abnormal luggage along with relevant information in a distributed manner. This facilitates subsequent operations such as accessing and viewing corresponding alarm information, tamper-proof verification, and data statistics by the edge computing device management server.

3 | VISUAL ALGORITHM DESIGN

3.1 | Object detection algorithm

Compared to traditional RFID luggage tracking systems, the system designed in this paper utilizes cameras as the acquisition layer. It integrates deep learning models to train on luggage images for object detection, thereby forming an inference layer. Finally, based on the inference results to recognize targets and output the detection, corresponding algorithms are designed to form the detection layer.

The inference layer in this paper is constructed using the YOLOv5 network architecture for object recognition training. The YOLOv5 network architecture is depicted in Figure 2:

In Figure 2, YOLOv5 utilizes CSPDarknet53 as its backbone network. CSPDarknet53 is a deep convolutional neural network designed for extracting features from images. Following the backbone, YOLOv5 introduces a feature pyramid network (FPN) structure in the neck portion of the network, which merges feature maps from different levels to better accommodate targets of various sizes. The head of the YOLOv5 network comprises several detection layers responsible for predicting bounding boxes of different sizes. Each detection layer is tasked with predicting the position and category of a set of anchor boxes. Additionally, YOLOv5 employs anchor boxes of varying sizes to adapt to targets of different dimensions and shapes. These anchor boxes are automatically adjusted during the training process to enhance detection accuracy. For activation functions, YOLOv5 uses Leaky ReLU, and it employs the complete intersection over union (CIoU) loss function to optimize detection performance.

Based on a dataset of over 4600 original images, data annotation and augmentation were performed, including geometric and pixel-level augmentation [30, 31]. These augmentations encompassed five categories of luggage, including soft bags and hard cases, among others. The labelling was classified accordingly. The test results of the luggage transport line at a certain airport are depicted in Figure 3.

Figure 3 shows surveillance cameras monitoring various luggage transport lines at a certain airport. The classification numbers correspond to: hard case—0, cardboard box—1, cardboard box with packing—2, soft bag—3, soft bag with packing—4. The testing time was 2 h, and the corresponding detection accuracy is shown in Table 1.



FIGURE 3 Test set validation results.

TABLE 1 Accuracy test table for the model.

Number of target categories	Accuracy
Hard case or Hard-shell luggage	93.3%
Cardboard box or paper box	95.4%
Cardboard box with framing	97.6%
Soft package	90.6%
Soft package + box	92.7%
Average	93.9%

The accuracy calculation formula is

$$\text{Accuracy} = \frac{\text{The total quantity detected in 2 h}}{\text{The total quantity of luggage detected in 2 h}}$$

3.2 | Luggage anomaly detection algorithm

Based on target detection, an algorithm is designed to detect luggage retention situations, primarily focusing on detecting cases where luggage falls off the conveyor belt and gets stuck. The algorithm flow chart is depicted in Figure 4:

The algorithm flow mainly consists of two steps:

- Step 1: Primarily involves executing the inference framework to identify and classify luggage, outputting luggage corner information and classification details.
- Step 2: By defining the ROI (region of Interest) area (the trajectory area where luggage normally moves on the conveyor belt), monitor whether there is any boundary-crossing behaviour of luggage within this region. If such behaviour is detected, start frame counting. If, after a set number of

frames, there is still a luggage target outside the ROI with the same classification, it is determined that there is luggage anomaly retention. This triggers an alarm and outputs data such as the current frame, time, camera ID, luggage type ID, etc.

For defining the ROI (region of Interest), two approaches were considered during the initial algorithm design phase.

- Approach 1: Defining the area outside the luggage conveyor belt as the ROI and detecting any luggage entering this region. If luggage is detected in this area, it indicates luggage anomaly retention.
- Approach 2: Similar to the above approach.

However, during actual testing, it was observed that some camera views were spacious. In the process of object tracking, Approach 1 resulted in a larger ROI area compared to Approach 2. Therefore, a comparative experiment was conducted using five cameras at different locations to compare the performance of Approach 1 and Approach 2. The comparison results are shown in Table 2.

3.3 | Perspective correction algorithm

The algorithm uses an improved ray casting method to correct the perspective differences caused by the varying sizes of objects in the field of view, particularly the issue of objects appearing smaller in the distance and larger when closer to the

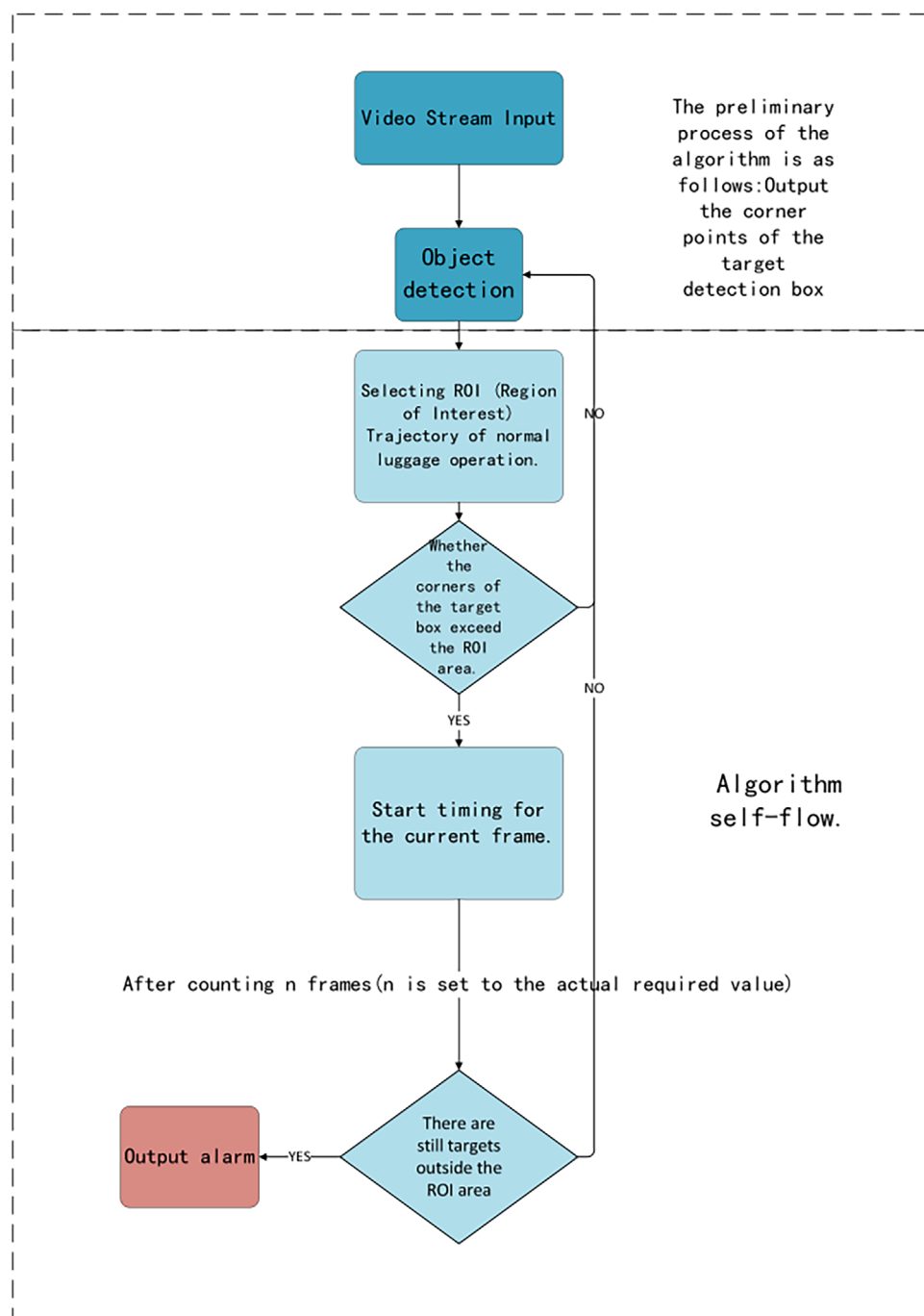


FIGURE 4 Algorithm flowchart.

TABLE 2 Performance comparison table of methods.

Name	Method 1	Method 2
Average area ratio	62.4%	51.8%
Single-frame processing time	52.3 ms	44.6 ms

camera. This correction is crucial, especially when objects closer to the camera obstruct the ROI region. The algorithm diagram is depicted in Figure 5.

As shown in the diagram above, the green area represents the conveyor belt area, with the arrows indicating the direc-

tion of movement. The blue box denotes the selected ROI region, which is the area to be evaluated. The black boxes represent the detected bounding boxes of luggage targets, with A, B, C, and D denoting the four corner points inferred from the object detection. The illustration includes two examples of luggage, labelled as X and Y. The algorithmic procedure is as follows:

- From each of the four corner points A, B, C, and D of any luggage item, rays are drawn in any direction. In the diagram, two types of rays are depicted: one perpendicular to

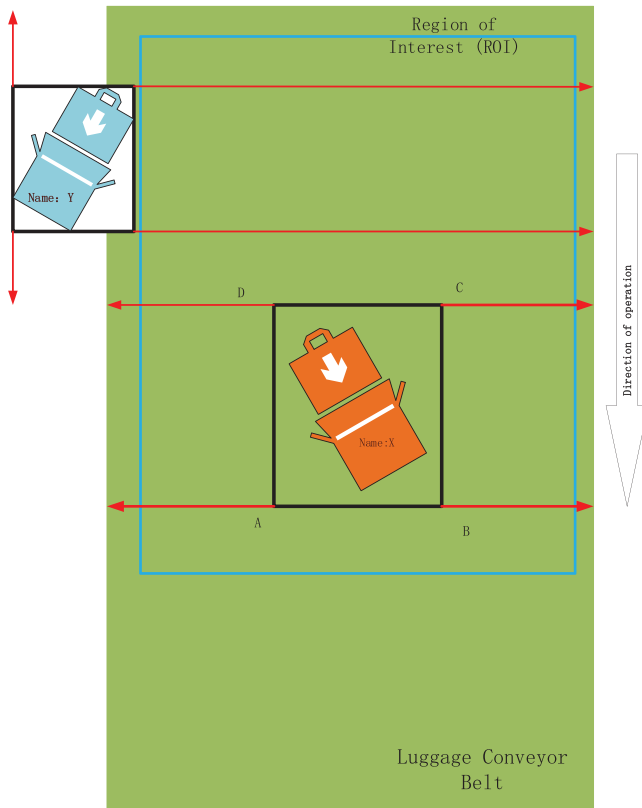


FIGURE 5 Improved ray casting schematic.

the direction of luggage movement and another parallel to the direction of luggage movement.

- Check if the polygon defining the ROI region has at least three vertices because a shape with fewer than three vertices cannot be considered a polygon. If there are fewer than three vertices, prompt a failure in ROI creation.
- Using point A as an example, the algorithm iterates through all the edges of the polygon and checks if each edge intersects with the ray. It then counts the number of intersection points between the ray and the edges of the ROI polygon. If the number of intersection points is odd, it indicates that the corner point is inside the ROI region. Otherwise, the corner point is outside the ROI region.
- To determine whether a piece of luggage has strayed out of bounds and is therefore considered to be abnormally retained, the algorithm checks if all four corner points (A, B, C, D) of the luggage are outside the ROI area. If all four points are not inside the ROI region, the luggage is judged to have exhibited out-of-bounds behaviour, indicating abnormal retention (as shown by luggage Y in Figure 5). Otherwise, the luggage is considered to be normal (as shown by luggage X in Figure 5).

The above algorithm helps avoid the problem of luggage appearing disproportionately smaller or larger due to the camera's installation angle, thereby obscuring the ROI boundary.

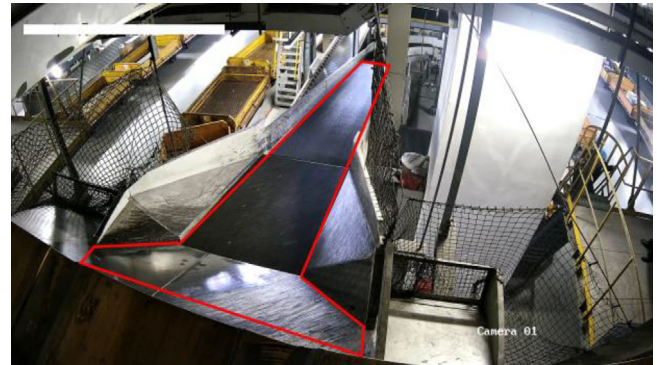


FIGURE 6 ROI region definition example.

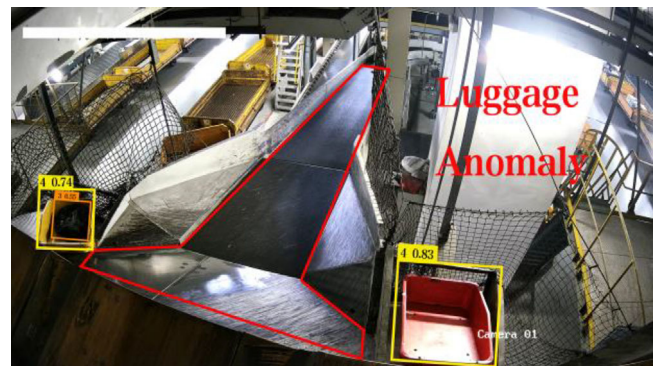


FIGURE 7 Algorithm detection results.

The test results with the ROI area selected are shown in Figure 6, and the final algorithm detection effect is presented in Figure 7. Since the training dataset did not include instances of luggage abnormal retention, the image depicts a test environment at an airport where luggage abnormal retention was artificially simulated.

4 | BLOCKCHAIN APPLICATIONS

4.1 | Blockchain framework

Existing data storage methods at airports primarily rely on traditional database systems, such as relational databases or distributed databases. Compared to blockchain, traditional databases have several disadvantages: they are usually managed by centralized servers or data centres, making the data vulnerable to attacks. If the server is attacked or fails, it could lead to data loss or leakage. Additionally, data can be altered by administrators or malicious attackers, and such alterations are difficult to trace. This poses risks in scenarios where data integrity is highly demanded. There are security risks associated with exposing vulnerabilities in airport operations and maintenance, which can be easily exploited by criminals.

In contrast, blockchain technology offers features such as decentralization, immutability, and transparency, providing higher levels of data security and trustworthiness. Through

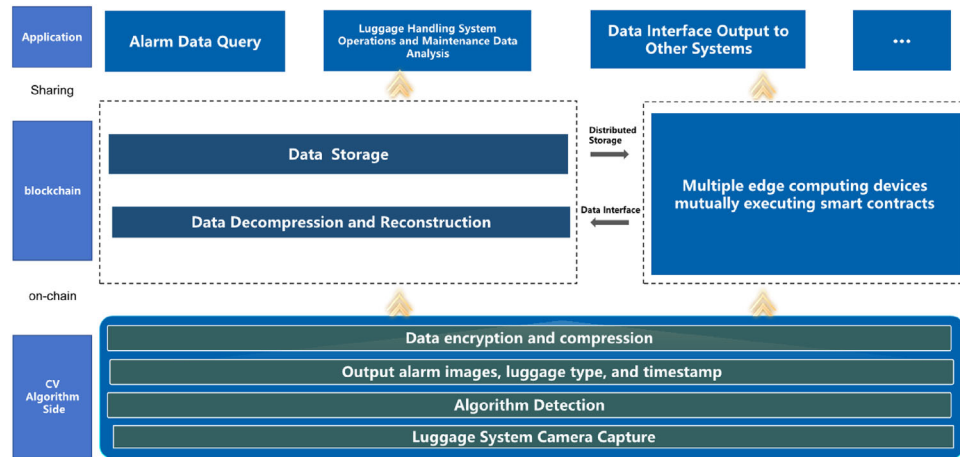


FIGURE 8 Storage framework and process diagram.

distributed storage, consensus mechanisms, and encryption algorithms, blockchain technology effectively addresses some of the issues present in traditional databases.

To address the shortcomings of traditional databases, the system described in this paper leverages a private chain formed by edge computing devices combined with on-site video data processing for alarm results. Multiple edge computing devices encrypt and store data on the chain, and then decrypt and share it through intermediate nodes, aimed at application by the edge computing device management server.

The blockchain alarm information storage framework and process designed in this paper are illustrated in Figure 8.

The entire blockchain data storage framework is illustrated in Figure 8, which mainly consists of the visual algorithm layer, storage and decompression layer, and application layer. The primary functions of each layer are as follows:

- **Visual algorithm layer:** This layer primarily consists of network cameras and edge servers, responsible for data acquisition and encryption for blockchain storage. Network cameras are deployed for real-time data capture and are located at the remote end of the network topology, without direct access to the nodes of the private chain. Edge servers possess computational and storage capabilities, managing the corresponding network cameras to perform tasks such as node authentication, visual algorithm processing, and alarm output. They encrypt and upload alarm results, timestamps, and luggage types to the blockchain.
- **Blockchain layer:** Comprised of all edge servers deployed on-site, this layer is responsible for handling the data from the abnormal luggage detection system. Given that the amount of information regarding the number of luggage anomalies (provided by a certain airport) does not impose significant pressure on a certain number of edge servers, multiple edge computing servers and edge computing device management servers can be organized into a Hyperledger Fabric private chain. This layer primarily executes smart contracts, accesses ledger data, and provides application programming interfaces

(APIs). It offers trustworthy data services to the application layer.

- **Application layer:** This layer caters to operations personnel for data analysis, intelligent maintenance of luggage handling systems, data requirements for airport smart operations, etc., providing trustworthy traceability to end-users [32, 33].

4.2 | Image encryption and anti-tampering algorithm

Blockchain security is one of the key factors for its success, and encryption algorithms form the foundation for ensuring the confidentiality, integrity, and authentication of blockchain data. For the entire process of outputting, storing, and authorizing alarm images from video surveillance, utilizing image encryption technology and blockchain technology to securely store alarm images ensures that monitoring images will not leak critical operational information of the airport (primarily to prevent leakage in places where luggage is prone to be detained), while also securely and traceably storing alarm images and information. Using blockchain technology to provide credentials for the integrity of alarm images and to verify whether off-chain alarm images have been correctly and unaltered. This paper will design a method based on hash encoding to use alarm images, types of luggage for abnormal retention, timestamps, and camera IDs as keys, encrypting the images using chaos and DNA encryption, and then decrypting the images through user access permissions to the keys.

In the civil aviation industry, regulations and compliance requirements impose strict requirements on the handling of sensitive information. Image encryption can help organizations meet these compliance requirements and avoid potential security issues. The encryption algorithm used in this paper builds upon the chaos and DNA image encryption algorithm [34], incorporating improvements based on the combination of Chen hyperchaotic system, logistic chaotic system, and DNA encoding and decoding for colour image encryption. This

Algorithm 1 Image encryption algorithm

Input: Original image I , block size t
Output: Encrypted image Q
 SplitChannels(I);
 ZeroPadding(I , t);
 $p \leftarrow \text{GenerateLogisticChaoticSequence}()$;
 $R \leftarrow \text{MatrixTransformation}(p, M, N)$;
 $(X, Y, Z, H) \leftarrow \text{SolveSuperChaoticSequence}(R)$;
 $Q \leftarrow \text{DNACoding}(X, Y, Z, H)$;
return Q ;

encryption algorithm combines the advantages of all three [35–37], resulting in a system with large key capacity, strong noise resistance, robustness against cropping, and resistance to attacks. Compared to other existing algorithms, this algorithm increases the complexity of the algorithm and enhances the security of encryption. While ensuring that the encryption time delay is relatively small, it ensures the security performance of image encryption. The algorithm implementation involves the following key steps:

1. Capture the image information, split the image into three channels: R (red), G (green), and B (blue). Assign the rows and columns of the image to variables M and N , respectively, and then perform block division.
2. Zero-padding: Pad the rows and columns of the image so that they are divisible by the block size t . Obtain the new dimensions after zero-padding.
3. Generate Logistic chaotic sequence p .
4. Matrix transformation: Transform the sequence p into an integer within the range of 0 to 255, and convert it into an $M \times N$ two-dimensional matrix R .
5. Solve the five-dimensional hyper-chaotic sequence: Solve the five-dimensional hyper-chaotic system based on the matrix R , obtaining four chaotic sequences X , Y , Z , and H .
6. DNA encoding: Use X and Y to determine the DNA encoding method for I and R , respectively. Use Z to determine the operation method, while H represents the DNA decoding method. Encode each block of the original image in each channel according to the sequence number corresponding to X with DNA encoding, and encode each block according to the sequence number corresponding to Y with DNA encoding. Perform DNA operation on the blocks encoded earlier according to the sequence number corresponding to Z . Merge each block into a complete image Q .
7. Fuse the three channels of image Q to obtain the encrypted image.

Using the aforementioned algorithm 1, the image before encryption is shown in Figure 9, the detection alarm result images are encrypted, and the result is shown in Figure 10.

Histogram analysis was performed on the encrypted images. In the histogram, the horizontal axis ranges from 0 to 255, representing the frequency of occurrence of each data value in the entire image. Therefore, the distribution of pixel values in the image can be reflected in the histogram. As shown in

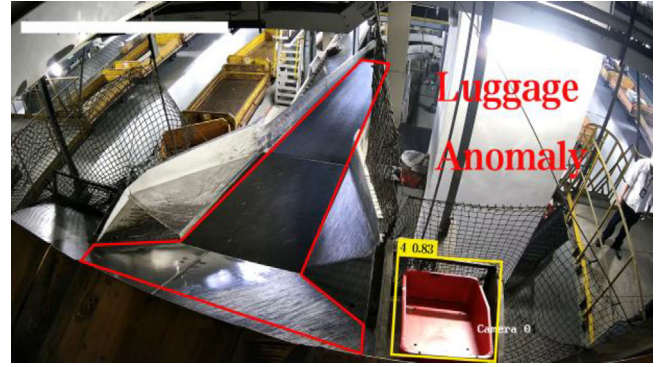


FIGURE 9 Alarm output image.



FIGURE 10 Encrypted image.

Figure 11, histograms of the R, G, and B channels before and after encryption are presented.

From Figure 11, it can be observed that the histograms of the three channels in the original image fluctuate irregularly, while the histograms of the three channels in the encrypted image are distributed flatly, exhibiting pseudo-randomness. This characteristic helps conceal the statistical properties of the original image, thereby effectively defending against large-scale histogram-based statistical attacks on the image.

The comparison of the image entropy after encryption indicates the level of disorder in pixel values. Its specific mathematical definition is given by the following formula:

$$H(x) = - \sum_{i=0}^{2^N-1} p(x_i) \log_2 p(x_i) \quad (1)$$

Where $p(x_i)$ represents the grayscale value in the image, and $p(x_i)$ represents its proportion. Here, 2^N represents the grayscale level of the image. If the grayscale level of an image is M , the maximum information entropy for an image, measured in bits per symbol, is given by: $H_{\max} = \log_2 M$. When $M = 256$, $H_{\max} = 8$. The closer it is to its maximum entropy, the more chaotic the ciphertext image is, indicating a better encryption effect. The table compares the entropy values of the three channels as shown in Table 3.

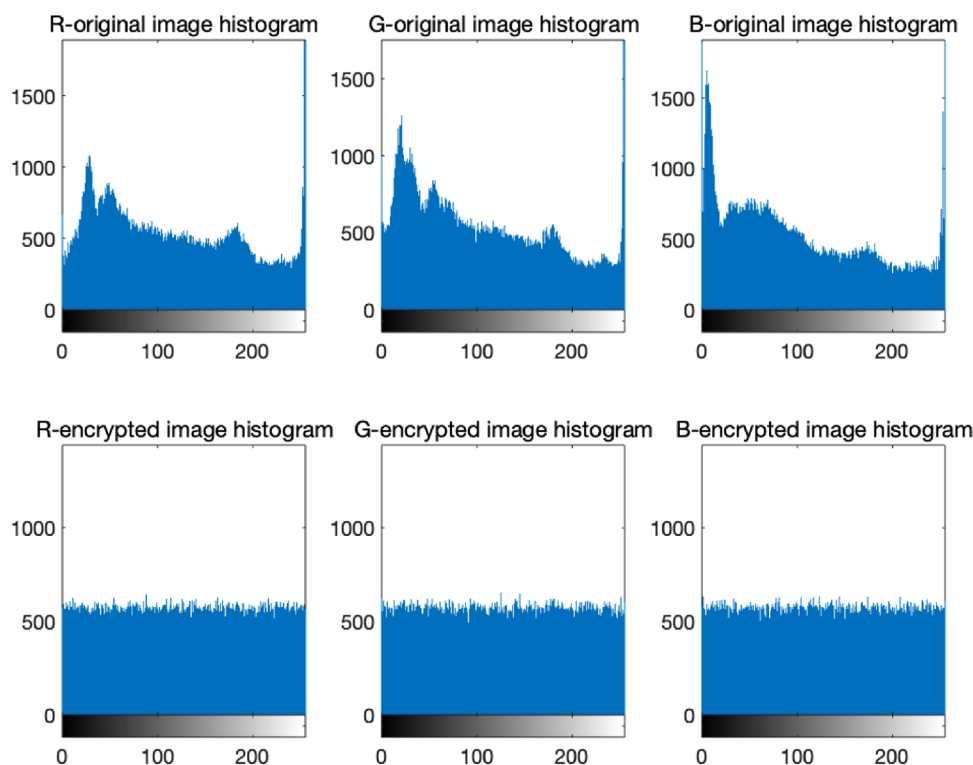


FIGURE 11 Histograms before and after encryption.

TABLE 3 Comparison of information entropy between original image and encrypted image.

	R-channel	G-channel	B-channel
Original image	7.7996	7.8174	7.7293
Encrypted image	7.999	7.9986	7.9987

As observed from Table 3, the ciphertext image approaches maximum entropy across all three channels, suggesting that the level of disorder in the image has reached its peak. Consequently, the encryption effect of the algorithm meets the anticipated standards.

The storage and usage of alarm image data face certain risks, even when encrypted. These risks include tampering due to human or non-human factors, as well as issues arising from storage mismanagement, leading to discrepancies between decrypted images and the actual ones. Smart contracts, also known as chaincodes, are vital components of the Hyperledger Fabric blockchain network. They execute various functional operations, such as storing and retrieving images and information. These chaincodes run in a Docker environment, stored and executed by supporting node processes. Designed to be independent, they do not affect the performance of other chaincodes in the network. Acting as carriers, chaincodes handle operations related to alarm image data. The primary data elements stored in the blockchain include image data, types of anomalous luggage, timestamps, and camera identifiers. These elements serve as credentials for determining whether

facial images in video surveillance have been tampered with, which is the core functionality of the anti-tampering algorithm. Therefore, leveraging the traceable and immutable nature of blockchain is essential to ensure the integrity of these elements.

When authorized users need to access original video images, they decrypt the image ciphertext to obtain the alarm image. Subsequently, they retrieve corresponding data from the blockchain to verify the integrity of the alarm image. Only if the verification is successful can the image be used to restore the original alarm image. Similarly, when it is necessary to verify whether the alarm image held by a user has been tampered with, the same steps need to be taken for verification. The anti-tampering process is illustrated in Figure 12.

5 | EXPERIMENTAL ANALYSIS

The experimental analysis was conducted using video stream analysis for alarm detection, with data collected at the baggage reclaim area of an airport in China. As the individual slot's luggage retention rate was not high, all instances of luggage retention during the experiment were artificially simulated. To fully consider the impact of lighting changes on detection, the experimental environment employed a 12-h video stream with artificial luggage retention created every half hour. The equipment used included Hikvision DS-2CD7A8XYZUV-WLS network cameras, with a main stream resolution of 3840×2160 and a frame rate of 25FPS. Edge servers 1, 2, and 3 were equipped with NVIDIA® Jetson AGX Xavier hardware, featuring integrated AGX XAVIER modules and running

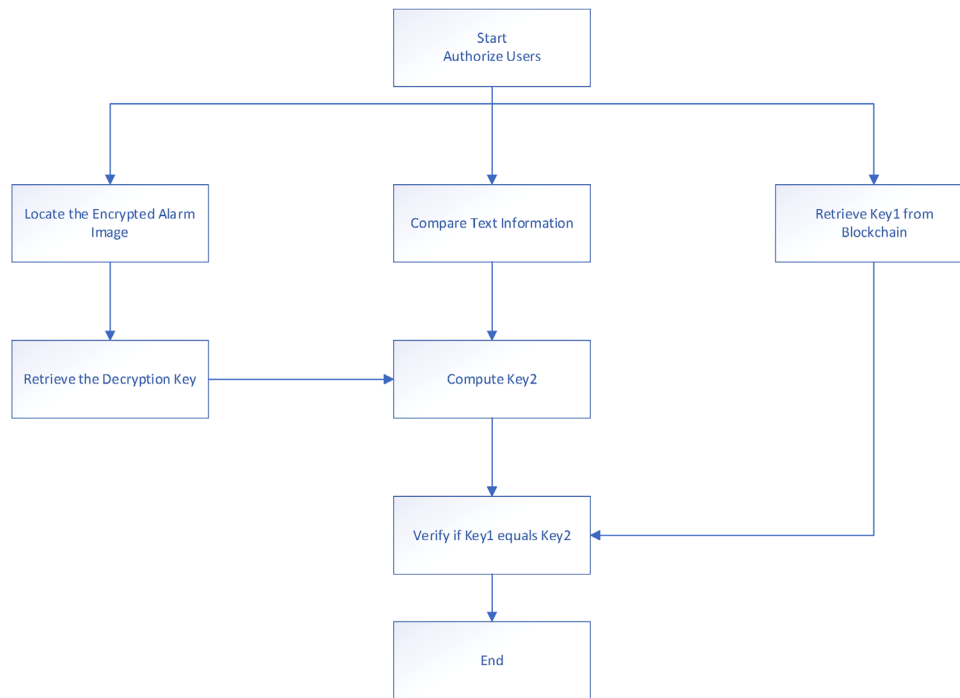


FIGURE 12 Tamper-proofing process flowchart.

TABLE 4 Visual algorithm function table.

	Detection literature [38]	Literature [39]	The algorithm in this paper
Delayed luggage detection	Yes	Yes	Yes
Supports complex scenarios	Yes	Yes	Yes
Utilizing neural networks	Yes	No	Yes
Detailed categorization of luggage	No	No	Yes
Edge computing processing	Unknown	Unknown	Yes

on the Ubuntu 18.04 operating system. Network bandwidth of 1000 Mbps. Testing resulted in a luggage recognition rate of 95.8% and a luggage retention detection rate of 96.9%. Additionally, a Hyperledger Fabric blockchain system was deployed in a Docker environment, with smart contracts written in Go language.

The results analysis mainly focuses on comparing the algorithm's functionality and the method's functionality.

5.1 | Visual detection algorithm

The functionality of the visual detection algorithm has been compared in accordance with Refs. [38, 39], as shown in Table 4.

For the existing issue of unclaimed luggage detection, this study utilized the YOLOv5 model for training and conducted a detailed classification of various types of luggage, so that staff could obtain more information for on-site handling. Meanwhile, by adopting an edge computing architecture to deploy the algo-

rithm, latency has been reduced, thereby enhancing the overall operational efficiency of the system. Moreover, the dataset was selected based on actual airport operations, making its detection accuracy more suited for airport scenarios.

In terms of image encryption algorithms, as a comparative experiment, Algorithm 1: the fractional-order Lorenz chaotic image encryption improved for the needs of this paper [40] and Algorithm 2: the Hyperchaotic image encryption algorithm based on dynamic filtering and joint permutation-diffusion improved for the needs of this paper [41] were selected. The comparison was made based on the original image encryption time, image decryption, and anti-tampering verification time, with the anti-tampering verification algorithm proposed in this paper as an example. The final time comparison is shown in Figure 13.

The encryption, decryption, and anti-tampering algorithms proposed in this paper may be slightly slower than Algorithm 1 while ensuring data security. However, they are relatively faster than Algorithm 2 due to their lesser complexity. Moreover, in terms of information entropy, Algorithm 1 does not perform as well as the algorithm used in this paper. As analysed in Section 4 of this paper, the encryption algorithm proposed in this paper meets the requirements for data security. In conclusion, the encryption and anti-tampering algorithms used in this paper can meet the practical requirements of airport usage.

In terms of overall system functionality analysis, existing RFID luggage tracking systems (see refs. [5, 6]) and visual detection systems for stranded luggage (see ref. [42]) were selected for functional comparison, as shown in Table 5.

Compared to the current mainstream RFID tracking systems, the technological system studied in this paper can complement or supplement them. It achieves comprehensive scene

Algorithm 2 Tamper-proof algorithm

Input: Alarm image, type of abnormal luggage:x, timestamp:y, camera number:z

Output: Tampering verification result

1. with open(img, "rb") as f://Open the image file in binary mode
2. for byte_block in iter(lambda: f.read(4096), b''):

sha256_hash.update(byte_block)//Read the file content in chunks using an iterator, encode it with SHA-256, and update the hash object.
3. return image_hash //Return the hexadecimal representation of the hash value for the alarm image.
4. hash_text(x,y,z)//Loading type of abnormal luggage:x, timestamp:y, camera number:z
5. text_bytes = text_data.encode//Convert text to bytes
6. text_hash = hashlib.sha256(text_bytes).hexdigest()//Calculate the hash value of text data
7. Key = image_hash + text_hash//Concatenate the hash values of the image and text data to form a key.
- //Validate whether tampering has occurred
8. response_ = cli.chaincode_invoke(requestor = cli.get_user('org1.example.com','Admin'),channel_name = "mychannel",peers = ['peer0.org1.example.com'],args = ['storeHash',Key]//Invoke chaincode function to store
9. response_ = cli.query_block(requestor = cli.get_user('org1.example.com','Admin'),channel_name = "mychannel",peers = ['peer0.org1.example.com'],block_number = x)//Query blockchain and verify hash value
10. Key2 = hash_image_and_text(image_file, ext_data)//Repeat steps 1–7
11. Key1 = blockchain.search(Key)
12. if Key1 == Key2 Return True or False//Compare the stored hash value with the computed hash value. If they are equal, it has not been tampered with; if they are not equal, it has been tampered with.

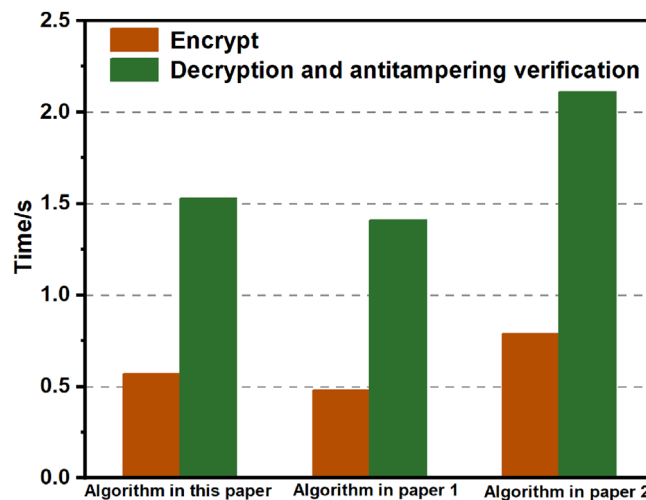
TABLE 5 System feature comparison table.

	Detection literature [5, 6]	Literature [44]	The system in this paper
Is it targeted towards luggage check-in scenarios	Yes	No	Yes
Can it provide localization	No	Yes	Yes
Real-time response	No	Yes	Yes
Information transmission security	Yes	No	Yes
Information storage security	No	No	Yes

positioning for monitoring stranded luggage. In comparison to existing visual detection systems for stranded luggage, the system designed in this paper offers advantages in terms of information storage and transmission security. Additionally, the algorithms developed are more suitable for monitoring stranded luggage in luggage handling systems.

6 | SUMMARY AND OUTLOOK

In summary, the technical solution proposed in this paper involves deploying multiple edge computing servers to con-

**FIGURE 13** Comparison of algorithm execution times.

struct a blockchain network. These servers are equipped with visual algorithms, image encryption algorithms, and anti-tampering algorithms. Network cameras capture images of the surroundings of the luggage conveyor system, and these images are transmitted to the edge computing servers for processing by the visual algorithms.

In testing, the visual detection algorithm achieved a luggage recognition rate of 96.9% and a luggage anomaly detection rate of 95.8% in the test videos. Real-time alerts for stranded luggage are transmitted to on-site displays for staff to handle promptly.

To ensure data security, luggage images are encrypted and stored on the blockchain. Authorized users or administrators can validate and extract the stored data, including images, categories, timestamps, and camera IDs, using the anti-tampering algorithm. Overall, this technology complements existing RFID luggage tracking systems by enabling real-time monitoring, alerting, and secure information transmission and storage of stranded luggage. Through experimental verification, it has the potential to enhance airport operational efficiency and information security.

However, the technical solution outlined in this paper has some shortcomings. The current dataset is insufficient to cover all types of luggage for detection, and further efforts are needed to increase the dataset and optimize the model to improve the detection rate. Additionally, the validation is limited to a single airport, and only airport personnel are informed.

In future research, upgrading the private chain to a consortium chain could be considered. During passenger travel, passengers' own information, such as flight numbers, names, and travel times, could be utilized as hash codes to obtain information about potential abnormal luggage retention. Corresponding images or data could be provided in the event of abnormal retention. Similarly, between airports, a consortium chain could be used to obtain information about abnormal luggage retention and notify arriving passengers of any anomalies regarding their luggage.

AUTHOR CONTRIBUTIONS

Yuzhou Chen: Conceptualization; methodology; software; validation; writing—original draft; writing—review and editing. **Gang Mao:** Formal analysis; data verification; software. **Xue Yang:** Conceptualization; resources and software; supervision; writing—review and editing. **Mingqian Du:** Writing—review and editing; supervision. **Hongqing Song:** Writing—review and editing; project administration.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Author elects to not share data: Due to the inclusion of proprietary commercial data and operational information from airlines in the dataset, we regret that we are unable to make this dataset publicly available. We hope for your understanding. Additionally, during the initial testing phase, we experimented with publicly available datasets online, such as the COCO dataset, and achieved comparable results.

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